M. Vidhya

192412151

Experiment 13: Car Price Prediction Model

Aim

To implement a **Car Price Prediction Model** using Python.

Python Program

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Sample Car Dataset
data = {
    'Year': [2014, 2013, 2017, 2011, 2018, 2015, 2016, 2012, 2019, 2020],
    'Present_Price': [5.59, 9.54, 9.85, 4.15, 6.87, 5.70, 8.12, 3.95, 10.00, 12.50],
    'Kms_Driven': [27000, 43000, 6900, 5200, 15000, 35000, 18000, 60000, 10000, 5000],
    'Fuel_Type': ['Petrol', 'Diesel', 'Petrol', 'Petrol', 'Diesel',
                  'Petrol', 'Diesel', 'Petrol', 'Diesel', 'Petrol'],
    'Seller_Type': ['Dealer', 'Dealer', 'Dealer', 'Individual', 'Dealer',
                    'Dealer', 'Dealer', 'Individual', 'Dealer', 'Dealer'],
    'Transmission': ['Manual', 'Manual', 'Manual', 'Manual', 'Automatic',
                     'Manual', 'Manual', 'Manual', 'Automatic', 'Automatic'],
    'Owner': [0, 0, 0, 0, 0, 0, 0, 1, 0, 0],
    'Selling_Price': [3.35, 4.75, 7.25, 2.85, 6.75, 4.60, 6.10, 1.95, 8.75, 11.25]
}

df = pd.DataFrame(data)

print("Car Dataset:")
print(df)

# Convert categorical columns into numbers
le = LabelEncoder()
```

```
df['Fuel_Type'] = le.fit_transform(df['Fuel_Type'])
df['Seller_Type'] = le.fit_transform(df['Seller_Type'])
df['Transmission'] = le.fit_transform(df['Transmission'])

# Features and target
X = df.drop('Selling_Price', axis=1)
y = df['Selling_Price']

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Create Linear Regression model
model = LinearRegression()

# Train model
model.fit(X_train, y_train)

# Predict test data
y_pred = model.predict(X_test)

print("\nActual Prices:")
print(y_test.values)

print("\nPredicted Prices:")
print(y_pred)

print("\nMean Squared Error:")
print(mean_squared_error(y_test, y_pred))

print("\nR2 Score:")
print(r2_score(y_test, y_pred))

# Predict new car price
new_car = pd.DataFrame({
    'Year': [2017],
    'Present_Price': [8.50],
    'Kms_Driven': [25000],
```

```
    'Fuel_Type': [1],       # Petrol=1, Diesel=0
    'Seller_Type': [0],     # Dealer=0
    'Transmission': [1],    # Manual=1, Automatic=0
    'Owner': [0]
})

prediction = model.predict(new_car)

print("\nPredicted Selling Price of New Car:")
print(prediction[0])
```

Sample Output

Actual Prices:

[8.75 4.75 4.6]

Predicted Prices:

[9.49922689 7.0210582 3.38437669]

Mean Squared Error:

2.3989287772437646

R2 Score:

0.3505682359695028

Predicted Selling Price of New Car:

5.917073476256633